Mme Doux
11th IB

Test 1

Name: .....

TIME: 50 minutes — : **31 marks**

GDC is allowed. Show all working.

EXERCISE 1 **[4 marks]**

A sequence is defined by $u_n = 15 - 4n$.

1. Find u_1 and u_2.

2. Write down a recursive formula for this sequence.

EXERCISE 2 **[2 marks]**

State whether the sequence $2, 6, 18, 54, \ldots$ is arithmetic, geometric, or neither. Justify your answer.

EXERCISE 3 **[5 marks]**

The third term of an arithmetic sequence is 16 and the ninth term is 46.

1. Find the first term and the common difference.

2. Find an expression for u_n, the n th term

3. Find the value of n for which $u_n = 151$.

EXERCISE 4 **[5 marks]**

A geometric sequence has first term $u_1 = 8$ and $u_4 = 216$.

1. Find the common ratio.

2. Find u_7.

3. Use your GDC to find the smallest value of n for which $u_n > 50\,000$.

EXERCISE 5 **[4 marks]**

Malia invests $5000 in an account that pays a nominal annual interest rate of 4%, compounded monthly.

1. Find the amount in the account after 5 years, giving your answer correct to 2 decimal places.

2. Find the minimum number of years for the investment to exceed $6500.

EXERCISE 6 **[3 marks]**

1. Write the series $3 + 7 + 11 + 15 + \cdots + 79$ using sigma notation.

2. Hence find the sum of this series.

EXERCISE 7 **[2 marks]**

Evaluate $\sum_{k=1}^{6} 5 \times 2^k$.

EXERCISE 8 **[4 marks]**

Consider the infinite geometric series $18 - 12 + 8 - \frac{16}{3} + \cdots$

1. Find the common ratio r.

2. Explain why this series converges.

3. Find the sum to infinity.

EXERCISE 9 **[2 marks]**

Write $0.\overline{45}$ as a fraction in lowest terms.

EXERCISE BONUS **[+5 marks]**

Two garden shops, Bloom and Petal, both open with a revenue of \$20 000 in their first month. Bloom's monthly revenue increases arithmetically by \$1500 each month. Petal's monthly revenue increases geometrically, multiplying by 1.05 each month.

Let B_n be Bloom's revenue in month n, and P_n be Petal's revenue in month n.

1. Show that $B_n = 1500n + 18\,500$.

2. Write down an expression for P_n.

3. Find the value of n for which Petal's revenue first exceeds Bloom's revenue.

4. Find Petal's total revenue over its first 12 months, giving your answer correct to the nearest dollar.

THE END